

CENG381 Stochastic Processes

M/M/1 and M/M/inf Queues - Answer Key 1

Q1 Solution

a) Traffic intensity and stability:

$$\rho = \lambda / \mu = 3/5 = 0.6$$

Since $\rho = 0.6 < 1$, the queue is STABLE.

$$P(\text{teller idle}) = \pi(0) = 1 - \rho = 1 - 0.6 = 0.4$$

=> the teller is idle 40% of the time.

$$P(\text{at least 2 customers}) = 1 - P(0) - P(1)$$

$$\pi(1) = (1-\rho) \cdot \rho^1 = 0.4 \cdot 0.6 = 0.24$$

$$P(N \geq 2) = 1 - 0.4 - 0.24 = 0.36$$

$$\text{Alternatively: } P(N \geq 2) = \sum_{n=2}^{\infty} (1-\rho) \cdot \rho^n = \rho^2 = 0.36$$

(using the tail of the geometric series).

b) E[N] and E[T] via Little's law:

$$E[N] = \rho / (1-\rho) = 0.6 / 0.4 = 1.5 \text{ customers on average}$$

$$\text{Little's law: } E[N] = \lambda \cdot E[T] \Rightarrow E[T] = E[N] / \lambda$$

$$E[T] = 1.5 / 3 = 0.5 \text{ hours} = 30 \text{ minutes}$$

c) Waiting time in queue E[T_q]:

$$E[T_q] = E[T] - 1/\mu = 0.5 - 1/5 = 0.5 - 0.2 = 0.3 \text{ hours} = 18 \text{ minutes}$$

Interpretation:

- Average time in service: $1/\mu = 0.2$ hrs (12 min), which is 40% of $E[T]$

- Average time waiting: $E[T_q] = 0.3$ hrs (18 min), which is 60% of $E[T]$

At $\rho = 0.6$, a typical customer spends more time waiting than being served.

As $\rho \rightarrow 1$, the waiting fraction $\rightarrow 100\%$ (queue explodes).

Q2 Solution

a) Normalisation of $\pi(n) = (1-\rho) \cdot \rho^n$:

$$\sum_{n=0}^{\infty} \pi(n) = \sum_{n=0}^{\infty} (1-\rho) \cdot \rho^n$$

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$$\begin{aligned} &= (1-\rho) * \sum_{n=0}^{\infty} \rho^n \\ &= (1-\rho) * 1/(1-\rho) \quad [\text{geometric series, } |\rho| < 1] \\ &= 1 \quad (\text{OK}) \end{aligned}$$

b) Derivation of $E[N] = \rho/(1-\rho)$:

$$\begin{aligned} E[N] &= \sum_{n=0}^{\infty} n * (1-\rho) * \rho^n \\ &= (1-\rho) * \sum_{n=0}^{\infty} n * \rho^n \end{aligned}$$

Use identity: $\sum_{n=0}^{\infty} n * x^n = x/(1-x)^2$ for $|x| < 1$

(Proof: start from $\sum x^n = 1/(1-x)$, differentiate both sides

w.r.t. x : $\sum n * x^{n-1} = 1/(1-x)^2$, multiply by x .)

$$\begin{aligned} \text{So: } E[N] &= (1-\rho) * \rho/(1-\rho)^2 \\ &= \rho / (1-\rho) \quad (\text{OK}) \end{aligned}$$

c) $E[N_q] = \rho^2/(1-\rho)$ and interpretation:

$$\begin{aligned} E[N] - E[N_q] &= \rho/(1-\rho) - \rho^2/(1-\rho) \\ &= (\rho - \rho^2) / (1-\rho) \\ &= \rho * (1-\rho) / (1-\rho) \\ &= \rho \end{aligned}$$

Interpretation: $E[N] - E[N_q] = \rho$ is the mean number of customers IN SERVICE (i.e. at the server, not queuing).

Why is this equal to ρ ?

The server is BUSY with probability $P(N \geq 1) = 1 - \pi(0) = 1 - (1-\rho) = \rho$.

Since there is only 1 server, the expected number in service at any time

$= P(\text{server busy}) * 1 = \rho$.

This is consistent: $E[N] = (\text{mean in queue}) + (\text{mean in service}) = E[N_q] + \rho$.

Q3 Solution

a) Stationary distribution for M/M/inf:

$$\rho = \lambda/\mu = 6/2 = 3$$

$$\pi(n) = e^{(-\rho)} * \rho^n / n! = e^{(-3)} * 3^n / n!$$

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This is the Poisson(3) distribution.

$$P(N = 3) = e^{(-3)} * 3^3 / 3!$$

$$= e^{(-3)} * 27 / 6$$

$$= e^{(-3)} * 4.5$$

$$e^{(-3)} \sim 0.04979$$

$$P(N = 3) \sim 0.04979 * 4.5 \sim 0.2240$$

About 22.4% chance of exactly 3 patients in the system.

b) E[N] and E[T] for M/M/inf:

$$E[N] = \rho = 3 \text{ (mean of Poisson(3))}$$

$$\text{Little's law: } E[T] = E[N] / \lambda = 3 / 6 = 0.5 \text{ hours} = 30 \text{ min}$$

This equals $1/\mu = 1/2$ hour exactly -- because in M/M/inf there is NO WAITING. Every patient immediately begins service, so the only time spent is the service time itself, with mean $1/\mu$.

c) Comparison: M/M/inf vs M/M/1 with fast server:

M/M/1 with $\lambda = 6$, $\mu = 12$ ($\rho = 6/12 = 0.5$):

$$E[N]_{\text{MM1}} = \rho / (1 - \rho) = 0.5 / 0.5 = 1 \text{ customer}$$

$$E[T]_{\text{MM1}} = E[N] / \lambda = 1/6 \text{ hours} \sim 10 \text{ minutes}$$

M/M/inf with $\lambda = 6$, $\mu = 2$ ($\rho = 3$):

$$E[T]_{\text{MMinf}} = 1/\mu = 1/2 \text{ hour} = 30 \text{ minutes}$$

Comparison:

$$E[T]_{\text{MM1}} = 10 \text{ min} < E[T]_{\text{MMinf}} = 30 \text{ min}$$

The M/M/1 with one fast doctor ($\mu=12$) gives much shorter system time.

Realism note: in a hospital, treatment time is determined by the patient's condition, NOT by staff capacity -- a doctor cannot simply treat patients 3x faster ($\mu=6 \rightarrow \mu=12$) by being alone. In reality:

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- M/M/inf better models a scenario where each patient needs their own dedicated specialist for a fixed treatment duration ($1/\mu = 30$ min).
- The 'fast single doctor' assumption may be unrealistic for emergencies.
- Trade-off: infinite servers = high staffing cost but no wait;
single fast server = low cost but requires doubling speed, which may not be medically feasible.